Module – 3

Transformers

&

Electrical Machines

Transformers – Power Generation

- **Till now** : Properties and concepts of magnetism and magnetic circuits
- Also, magnetic effect of electric circuits
- **Now :** Generation of EMF with help of magnetism

- EMF can be generated using :
  - Chemical action
  - Heating thermocouples, etc.

- Most popular : Electromagnetism

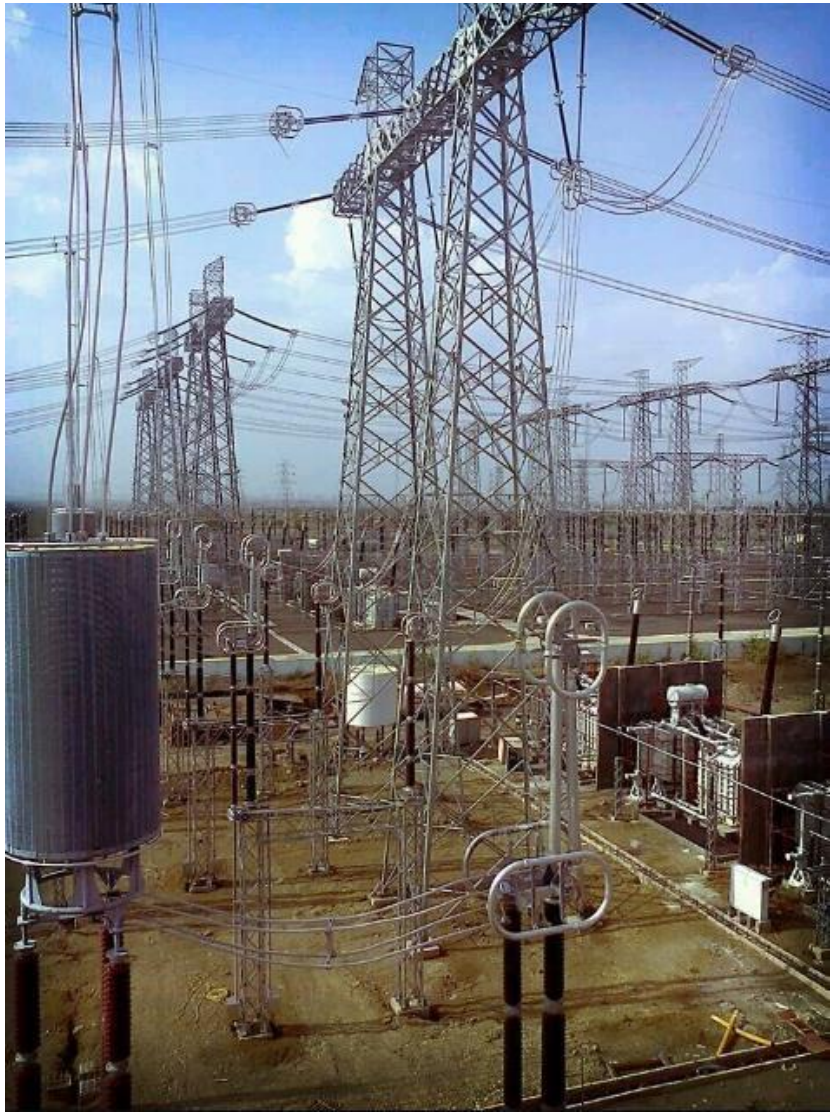
Transformers – Power Generation





Transformers – Power Transmission

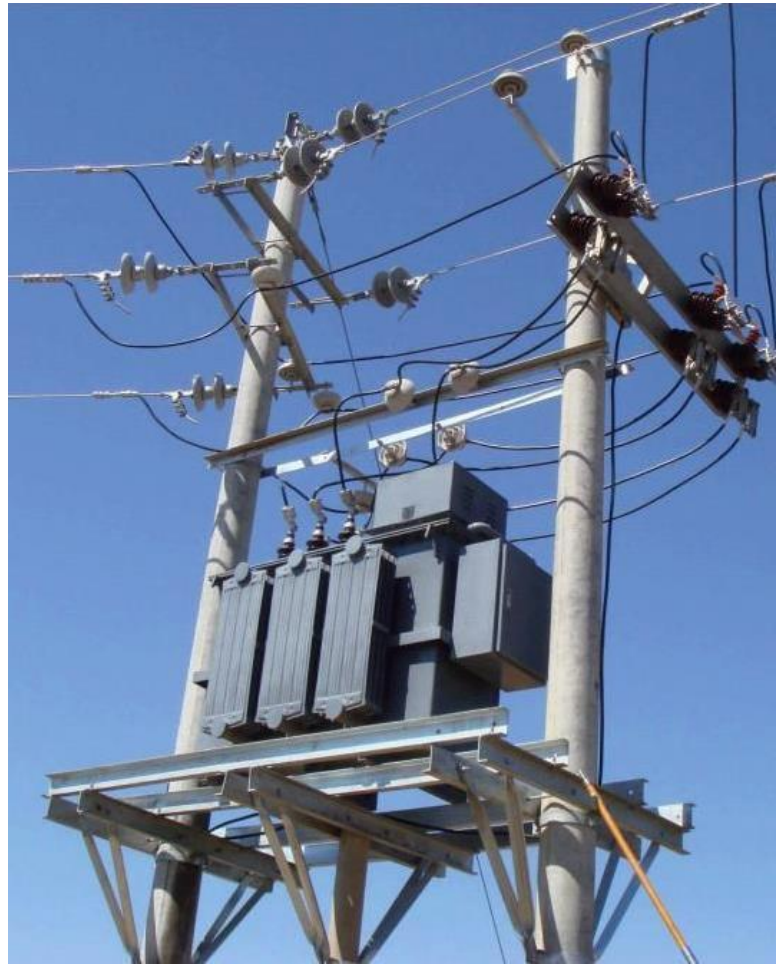




Transformers – Power Distribution



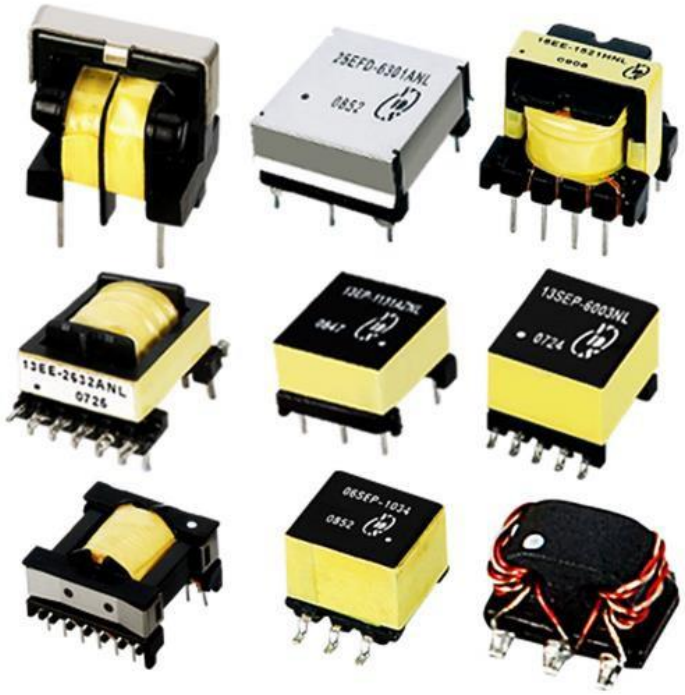


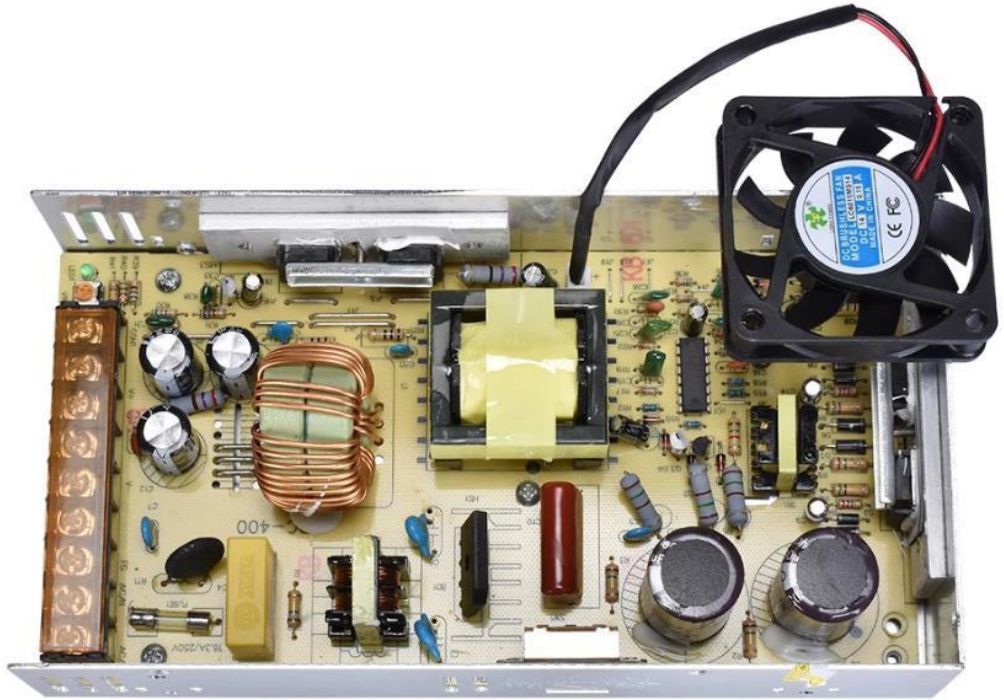


Transformers – Power Utilization



Welding transformer

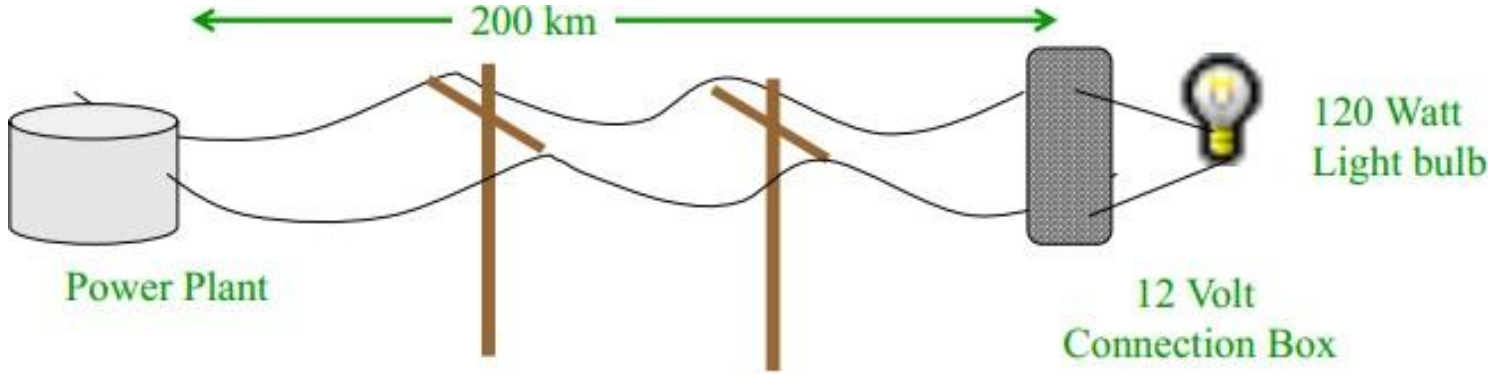




High frequency transformers used in electronic power supplies

NEED FOR TRANSFORMERS

Power transmission at high voltage while utilization at low voltage



- Estimating resistance of line, typically 0.001 Ω/m => 20 Ω for 200-km
- Current required to glow bulb: $I = P/V = 10A$
- Total power loss in line: $P_{loss} = I^2R = 10^2 \times 20 = 2000W$
- Efficiency: $\eta = \dfrac{P}{P_{loss} + P} = \dfrac{120}{2000 + 120} \times 100 = 5.66\%$ (very small!)

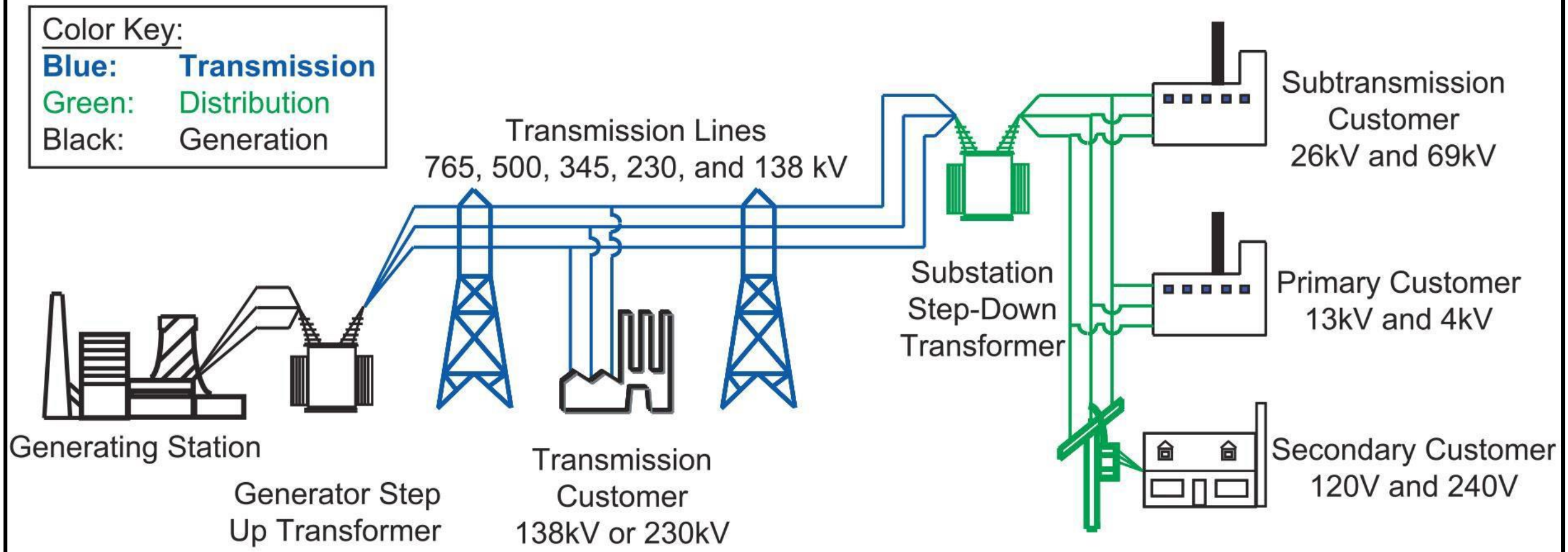
To **improve** η, decrease I in **transmission** (i.e. **increase** V for same power)

NEED FOR TRANSFORMERS

- **Transformer**: **Receives AC** electrical **power** at **one voltage** level and delivers/**transfers** it at **another voltage** level (**higher** for "**step-up**" and **lower** for "**step-down**").
- Need in modern power systems:
  - **Long distance power lines** can have **significant I^2R losses**.
  - The same power can be delivered by **high-voltage transmission** lines at a **fraction of the current** and with **much higher efficiency**.
  - This feature is one of the main **advantages of AC transmission** and distribution **over DC** system.

TRANSFORMERS IN POWER SYSTEM

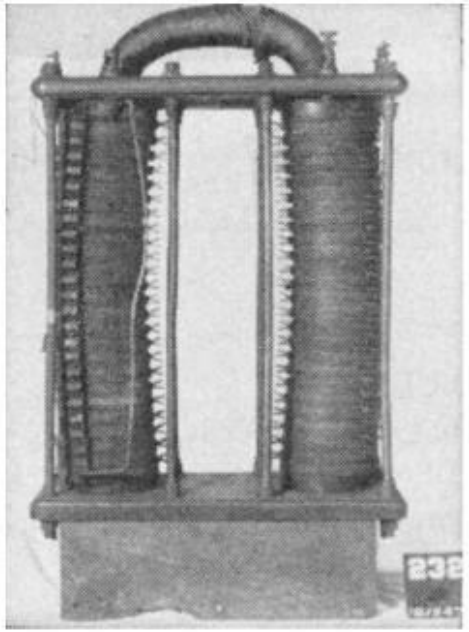
Step-Up Transformer at Generator and Step-Down Transformer at Consumer

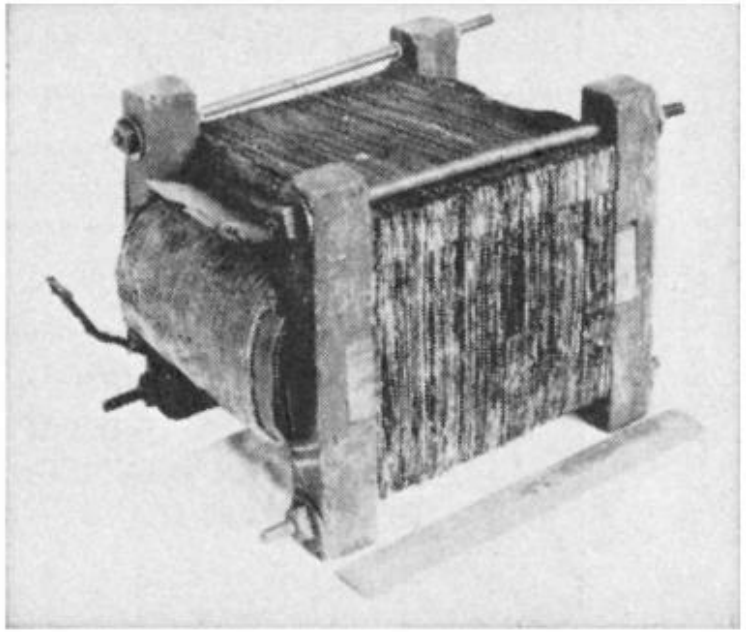


History and Invention

- 1886: Earliest known transformers used for first long-distance AC electric lighting system in Great Barrington, Massachusetts, USA.
- Used to transmit power from 25-hp steam-engine-driven alternator generating 500 V & 12 A to power light bulbs 4000-ft away.

Gaulard and Gibbs transformer





William Stanley's early transformer

SALIENT FEATURES OF TRANSFORMER

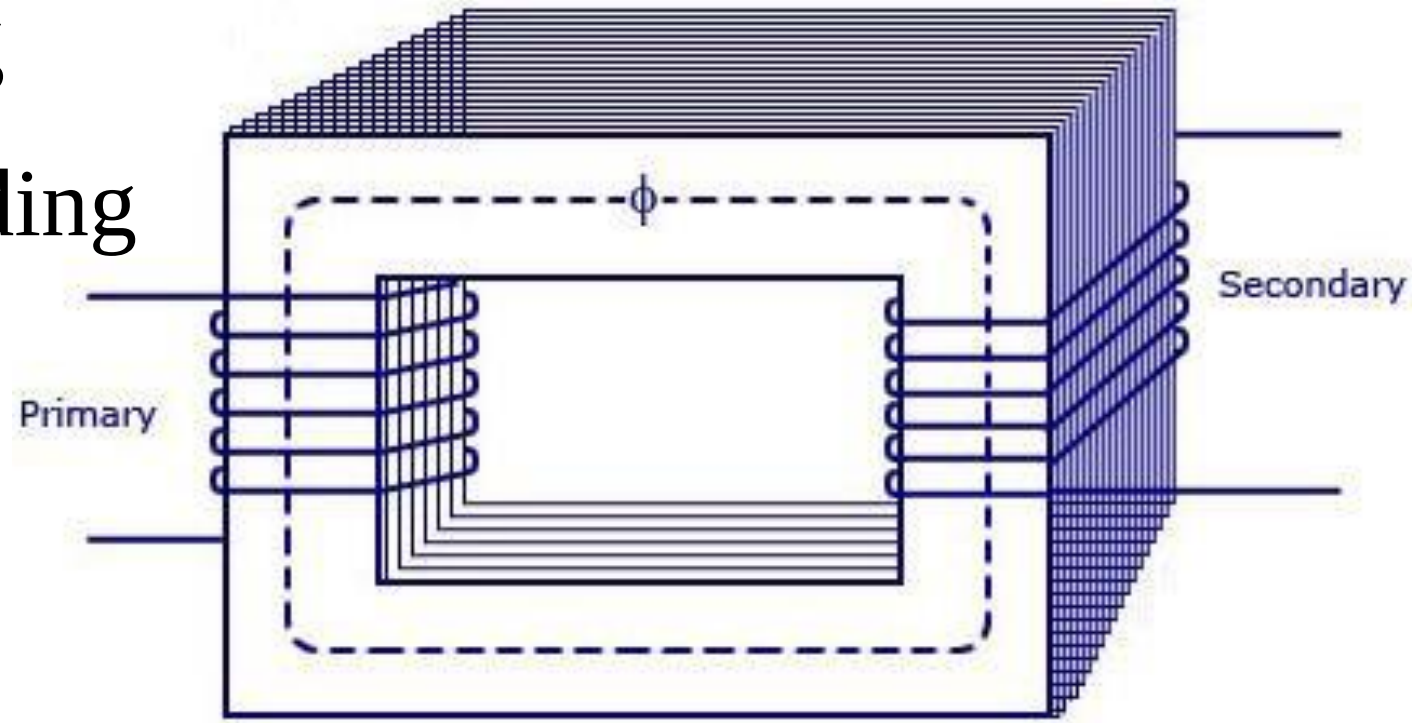
- **Transformation** from a certain (**rms**) **input voltage** (& current) value to a **different output voltage** (& current), keeping the AC signal **frequency constant**.
- Transformation occurs by means of common **magnetic field shared between two coils**.
- **No direct electrical connection** between input & output (exception: auto-transformers).
- Static device with no moving parts => **negligible maintenance**.
- **Highly efficient** power transfer (~98%) from input to output.

TRANSFORMER CONSTRUCTION

- Based on the phenomenon of **Mutual Inductance**.
- Consists of **two** electrical **coils** linked by a **common ferromagnetic core**.

➢ **Primary** (input) winding

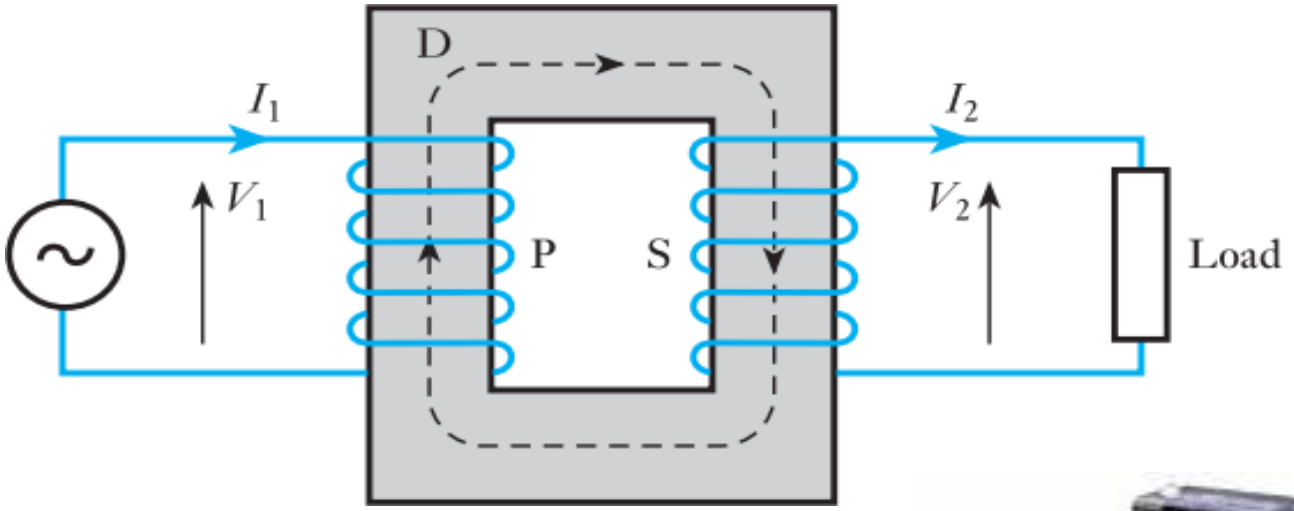
➢ **Secondary** (output) winding



TRANSFORMER CLASSIFICATION

- Based on **number of phases**:
  - **Single-phase transformer** (low power rating)
    Secondary (output) AC voltage often converted to DC for use in electronic applications.
  - **Three-phase transformer** (high power rating)
    Can be built as three single-phase transformer units or a single three-phase unit. Used in power systems/electrical grids

TRANSFORMER CLASSIFICATION

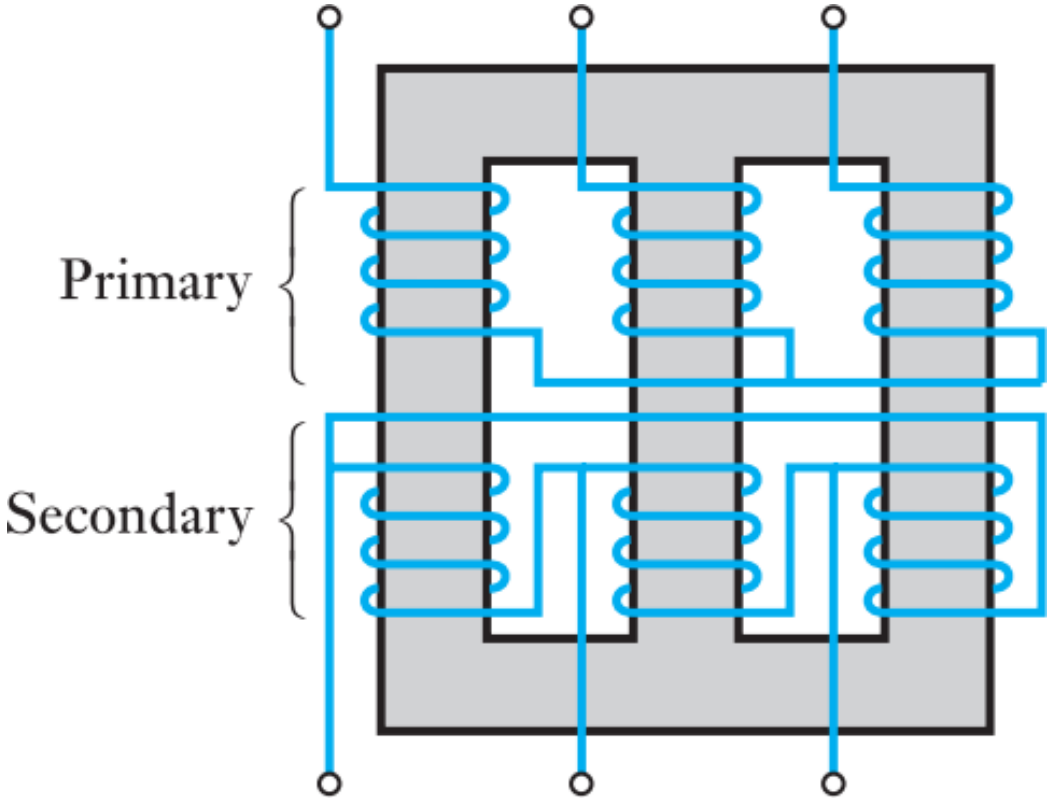


Single-phase transformer





TRANSFORMER CLASSIFICATION



Three-phase transformer





TRANSFORMER CLASSIFICATION

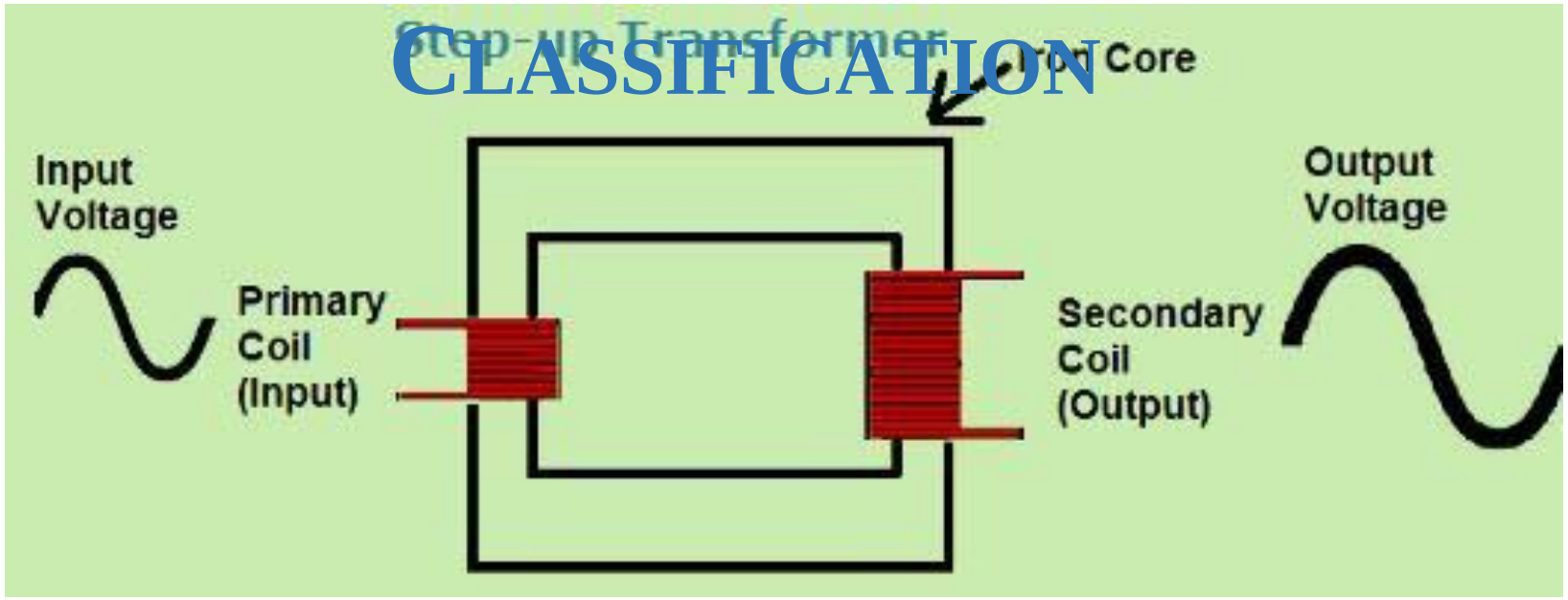
- Based on **voltage transformation level**:

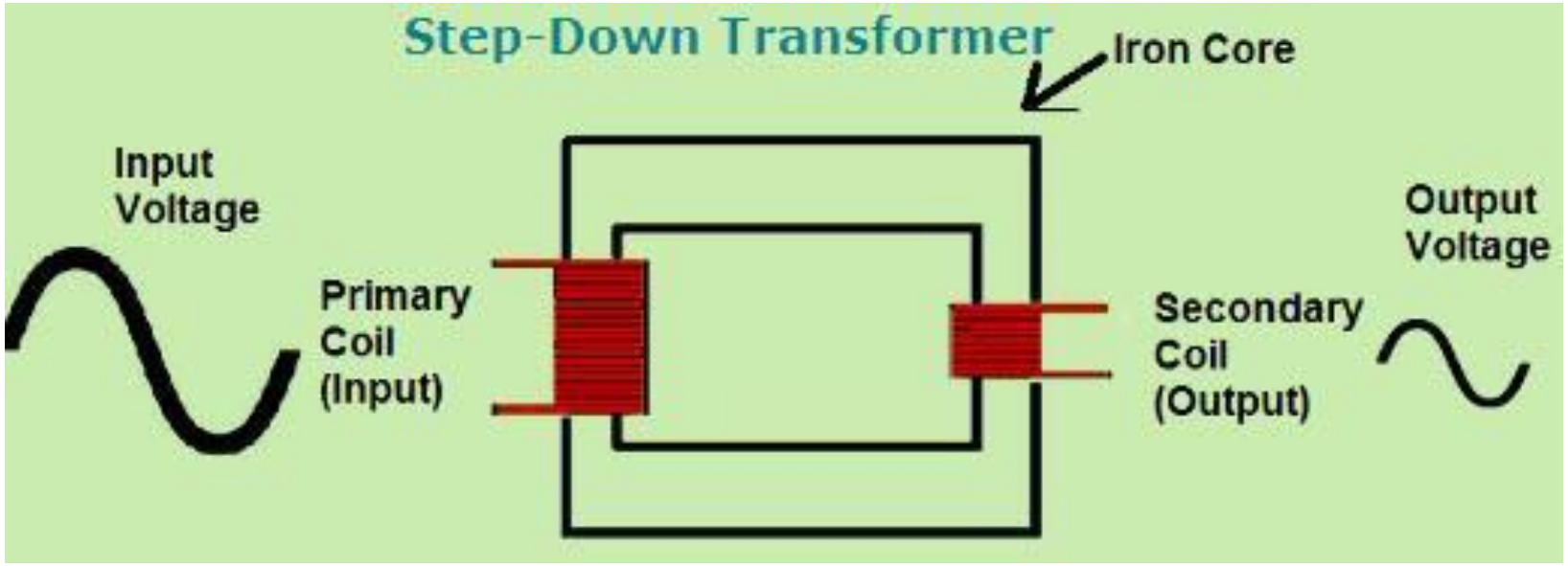
  - **Step-up transformer:**
    Secondary (output) voltage is “stepped up” (**magnified**) by a specific ratio **compared to primary** (input) voltage by using a **greater number of turns in secondary** winding.

  - **Step-down transformer:**
    Secondary (output) voltage is “stepped down” (**attenuated**) by a specific ratio **compared to primary** (input) voltage by using a **greater number of turns in primary** winding.

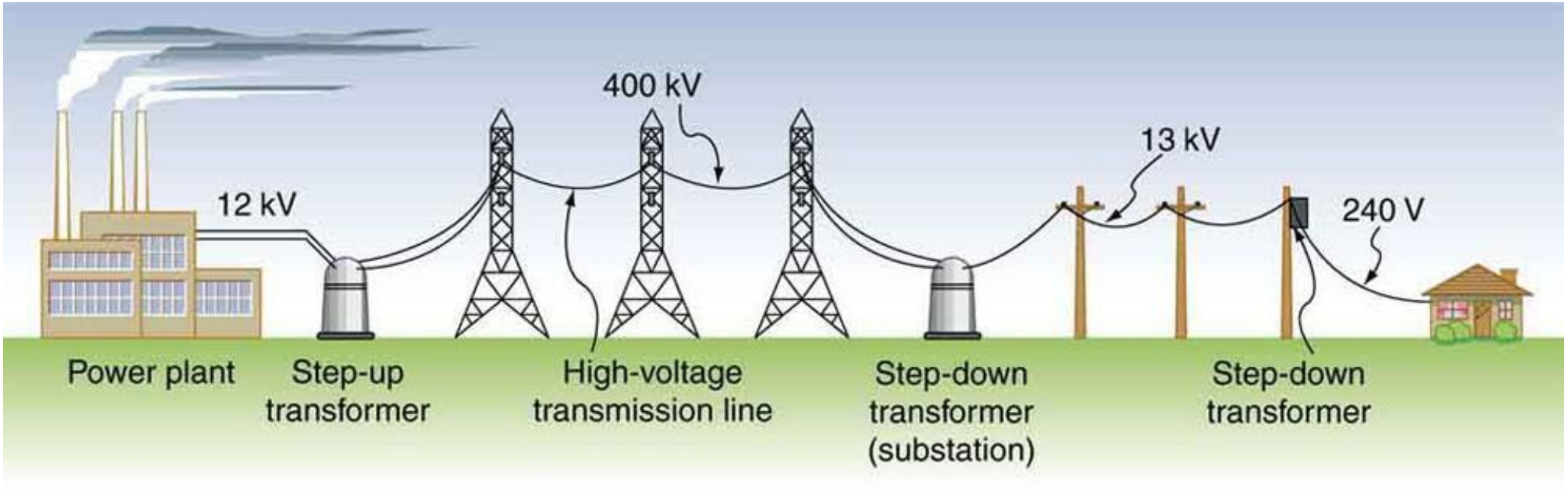
TRANSFORMER

CLASSIFICATION





TRANSFORMER CLASSIFICATION



Usage of Step-up and Step-down transformers in Power Systems

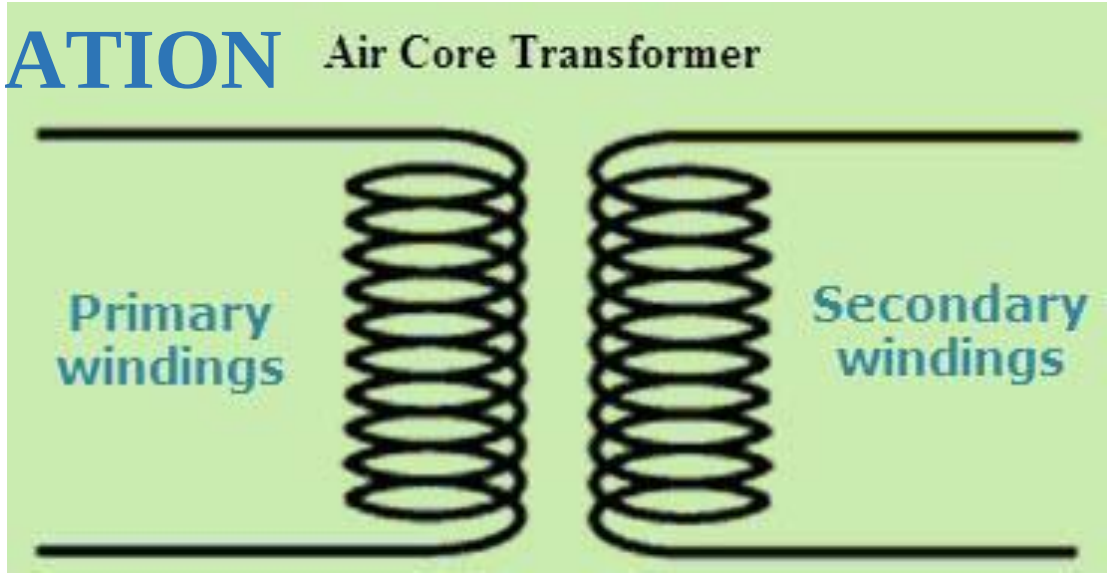
TRANSFORMER CLASSIFICATION

- Based on **core medium** used:
  - **Air core transformer:** Coils wound around hollow plastic or ceramic material ($\mu_r \sim 1$) => less flux linkage between coils. Used for **high frequency** applications (radio and television receivers).
  - **Iron / Steel core transformer:** High permeability ($\mu_r \sim 1500$) => mostly used in transformers in **power systems/electrical grids**. Susceptible to **Hysteresis loss** and **Eddy Current loss**.
  - **Ferrite core and other alloys** (e.g. AlNiCo): Used in **high frequency** transformers as they offer **reduced losses** in core.

TRANSFORMER CLASSIFICATION



Steel/Iron core transformer



Air core transformer



Ferrite core transformer

TRANSFORMER CLASSIFICATION

- Based on **usage in power system**:
  - **Power transformer:** Large rating transformers used in electrical **power transmission** networks for step up/down applications (e.g. **400 kV, 200 kV, 110 kV, 66 kV, 33kV**). Generally designed at operated at full load.
  - **Distribution transformer:** Comparatively small rating transformers used in **power distribution** networks (e.g. **11 kV, 6.6 kV, 3.3 kV, 440 V, 230 V**). Rarely operated at full load.

TRANSFORMER CLASSIFICATION





Power transformers

TRANSFORMER CLASSIFICATION







Distribution transformers

TRANSFORMER CLASSIFICATION

- Based on **application**:

  - **Measurement Transformers**

    1. **Current transformer** (CT) is used for **measuring current** in a circuit. **Primary** winding is connected in **series** with the **monitored circuit**.

    2. **Voltage transformer** (VT) is used for **measuring voltage** in a circuit. **Primary** winding is connected in **parallel** with the **monitored circuit**.

TRANSFORMER CLASSIFICATION

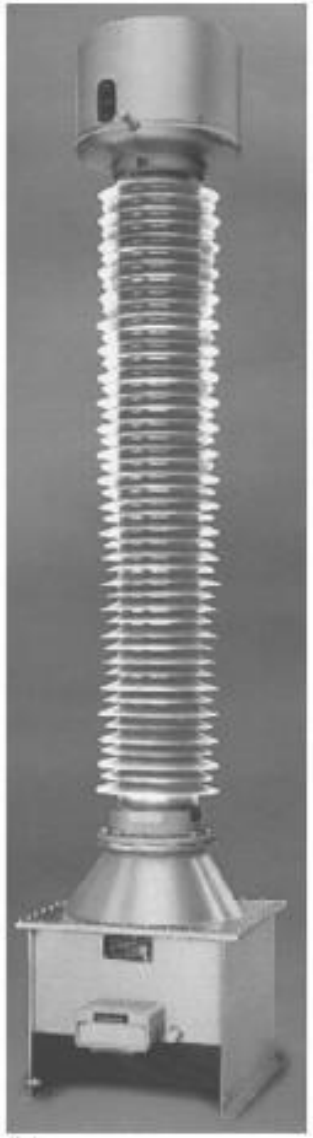
- Based on **application**:

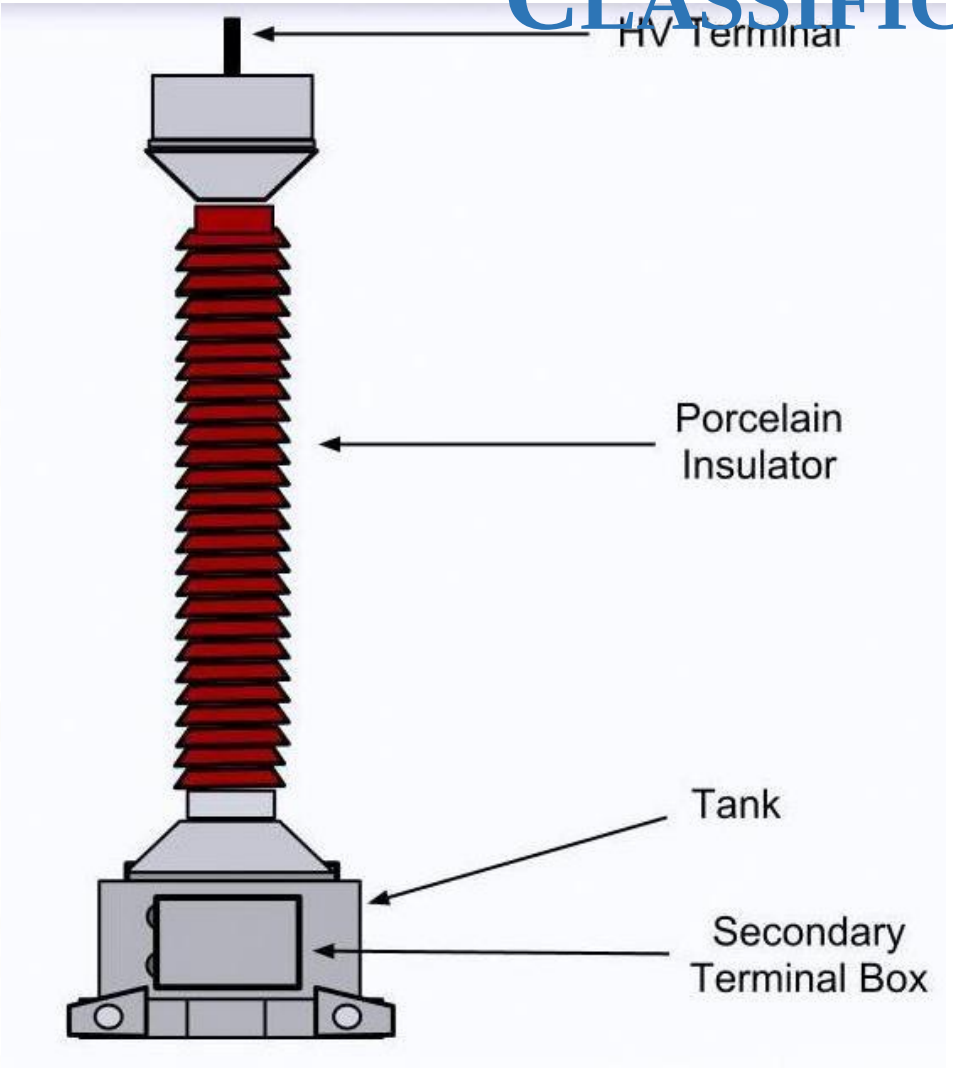
  - **Measurement Transformers**

    1. **Current transformer** (CT) is used for **measuring current** in a circuit. **Primary** winding is connected in **series** with the **monitored circuit**.

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TRANSFORMER CLASSIFICATION





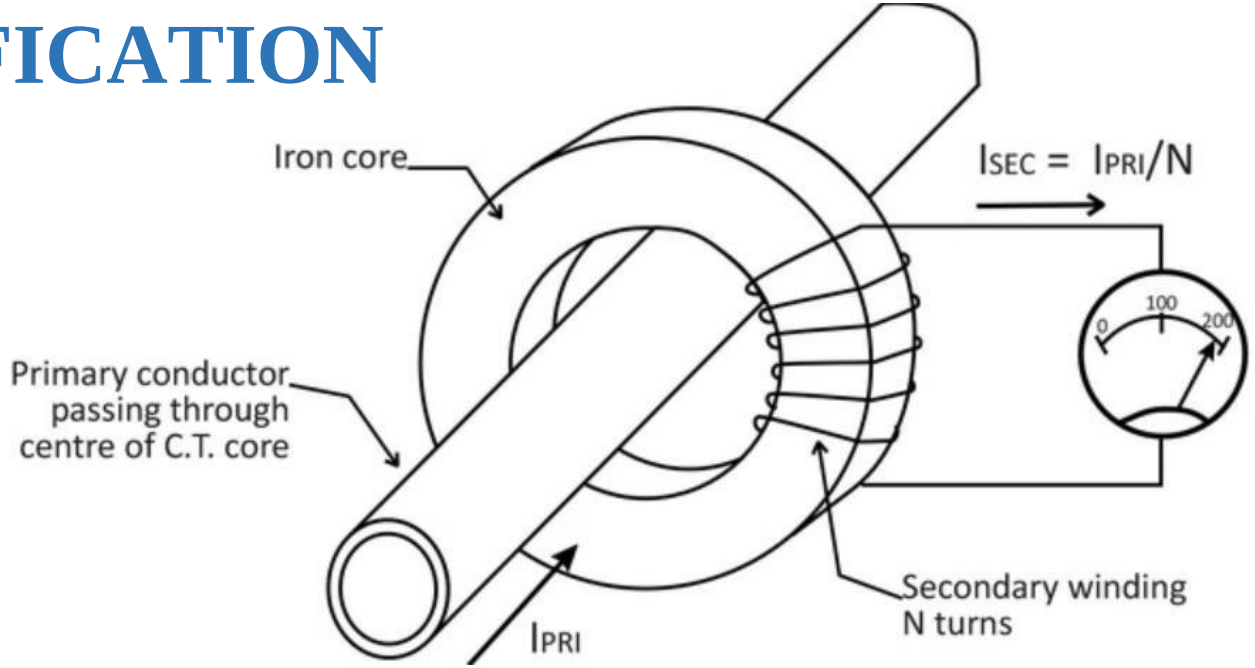


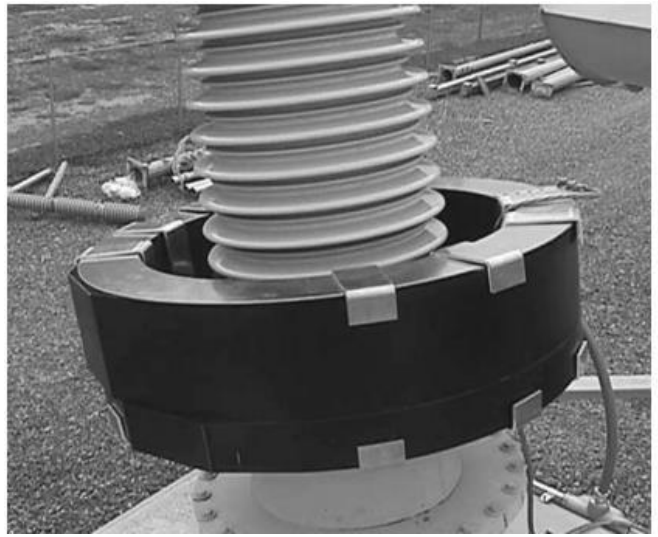
Voltage transformers

TRANSFORMER CLASSIFICATION

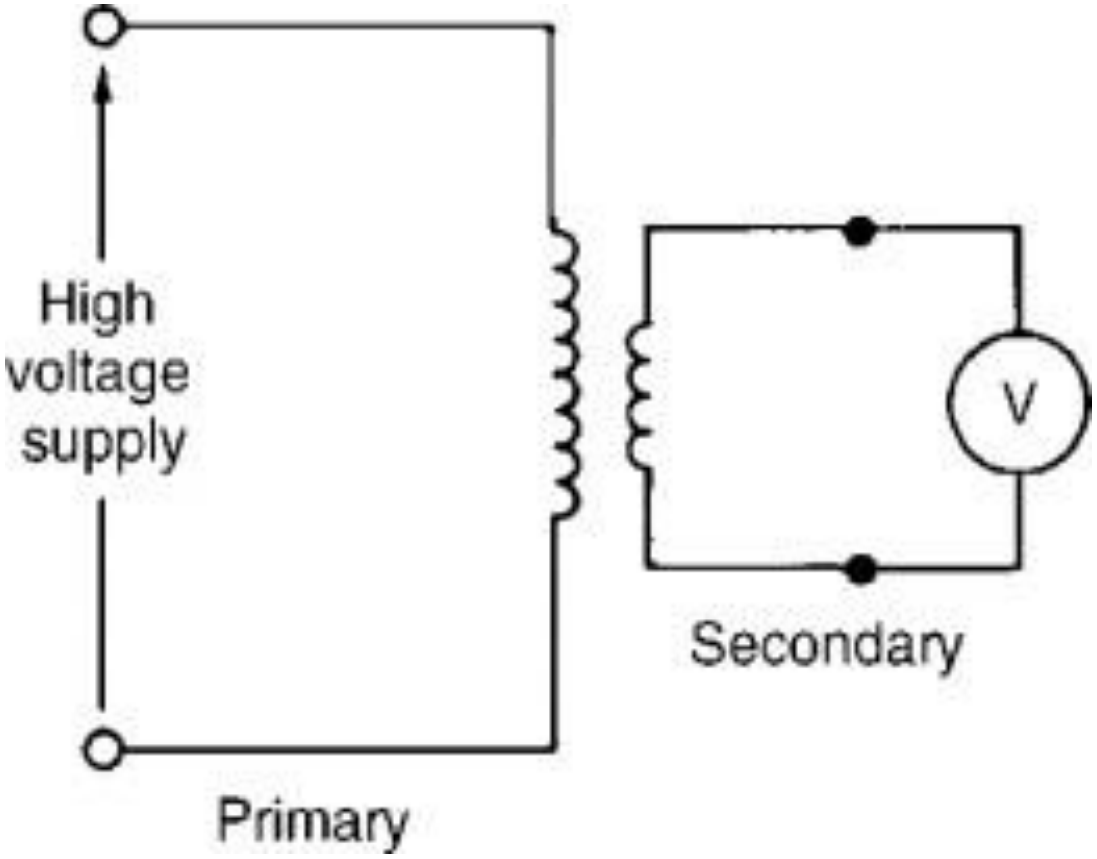


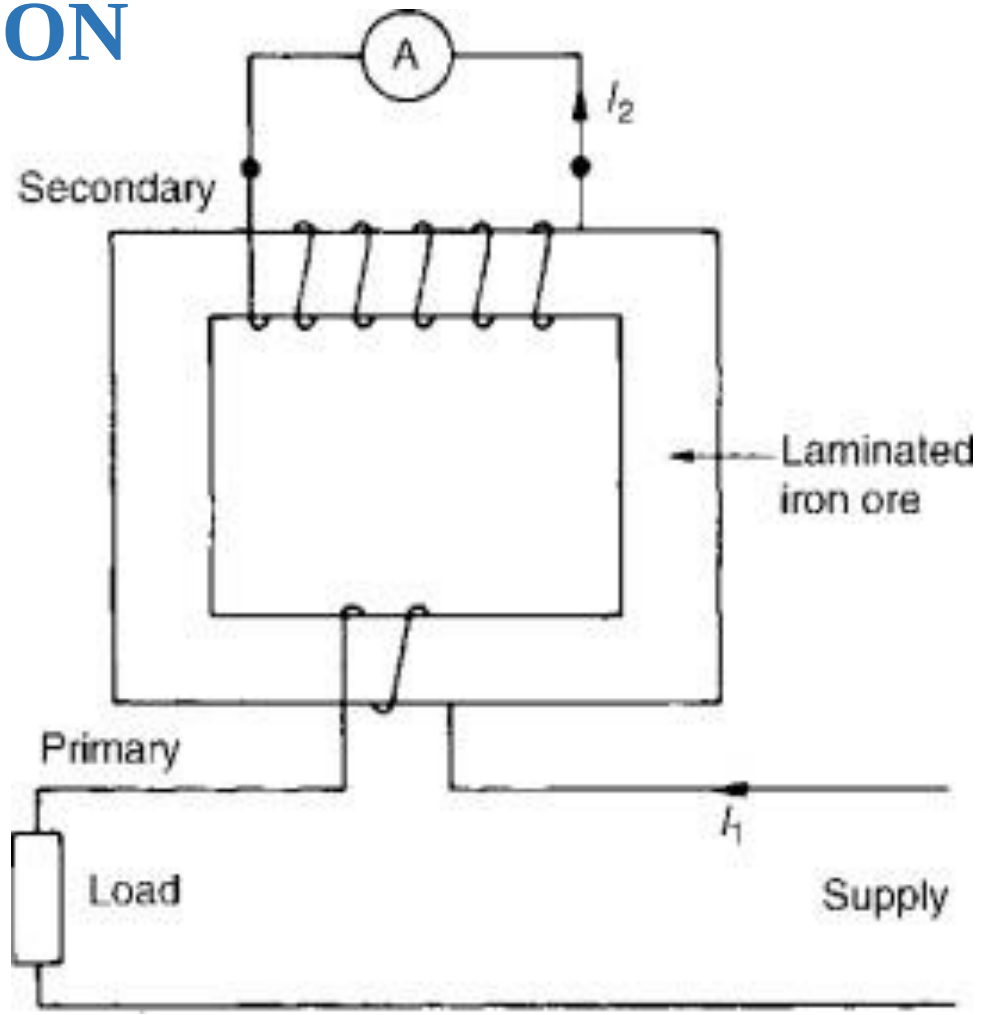
Current transformers





TRANSFORMER CLASSIFICATION





Simplified wiring diagram of VT (left) and CT (right)

TRANSFORMER CLASSIFICATION

- Based on **application**:
- **Isolation transformers:**
  - **Secondary voltage** is **equal to primary voltage** i.e. number of turns in both windings equal.
  - Used to **provide electrical isolation** and **protection against electric shock** e.g. bathroom shaver-sockets, portable electric tools, etc.
  - Also used to **suppress electrical noise** in sensitive devices e.g. mobile/wireless communication

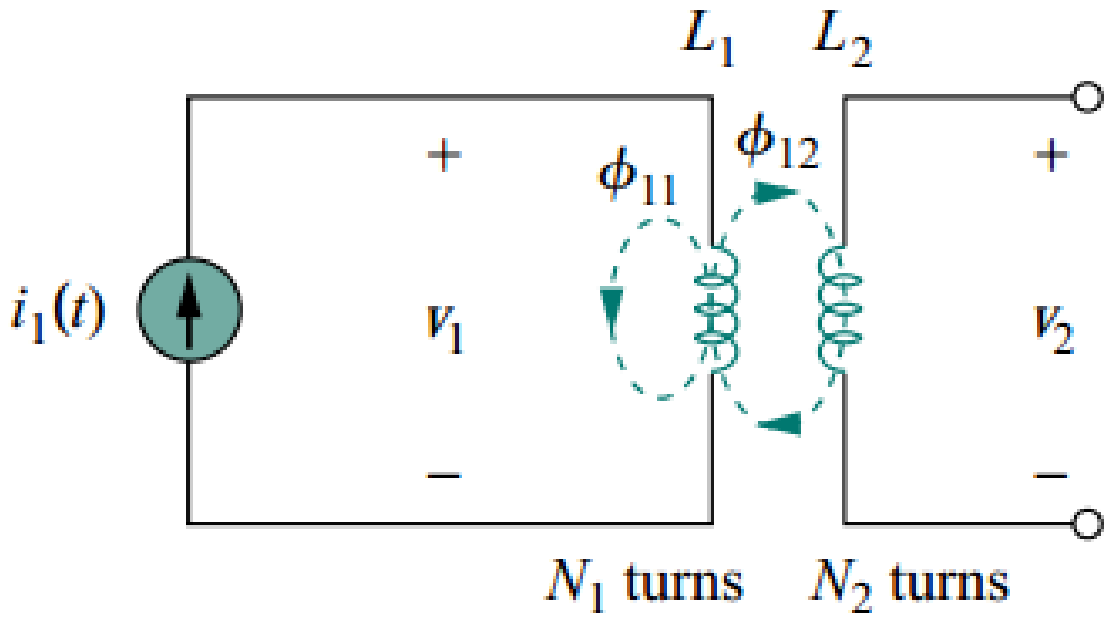
MAGNETICALLY COUPLED CIRCUITS

- **Transformer action** is based on **magnetic coupling**: **two** current carrying **coils** (with or without electrical contacts between them) affect each other through their **mutual magnetic field**.

- **Voltage induced in coil-1** (with coil-2 open circuit) can be written as,

$$\boxed{v_1} = N_1 \frac{d\phi_1}{dt} = N_1 \frac{d\phi_1}{dt}\frac{di_1}{di_1}$$

$$= \left(N_1 \frac{d\phi_1}{di_1} \right) \frac{di_1}{dt} = \boxed{L_1 \frac{di_1}{dt}}$$

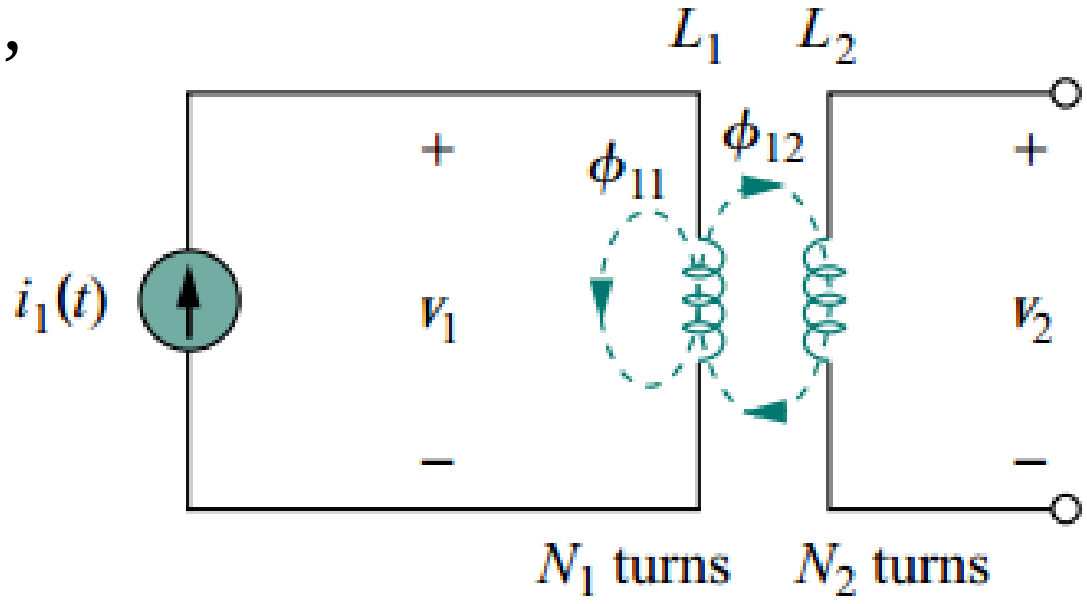


MAGNETICALLY COUPLED CIRCUITS

- **Voltage induced in coil-2** due to i_1 is,

$$\boxed{v_2} = N_2 \frac{d\phi_{12}}{dt} = N_2 \frac{d\phi_{12}}{dt}\frac{di_1}{di_1}$$

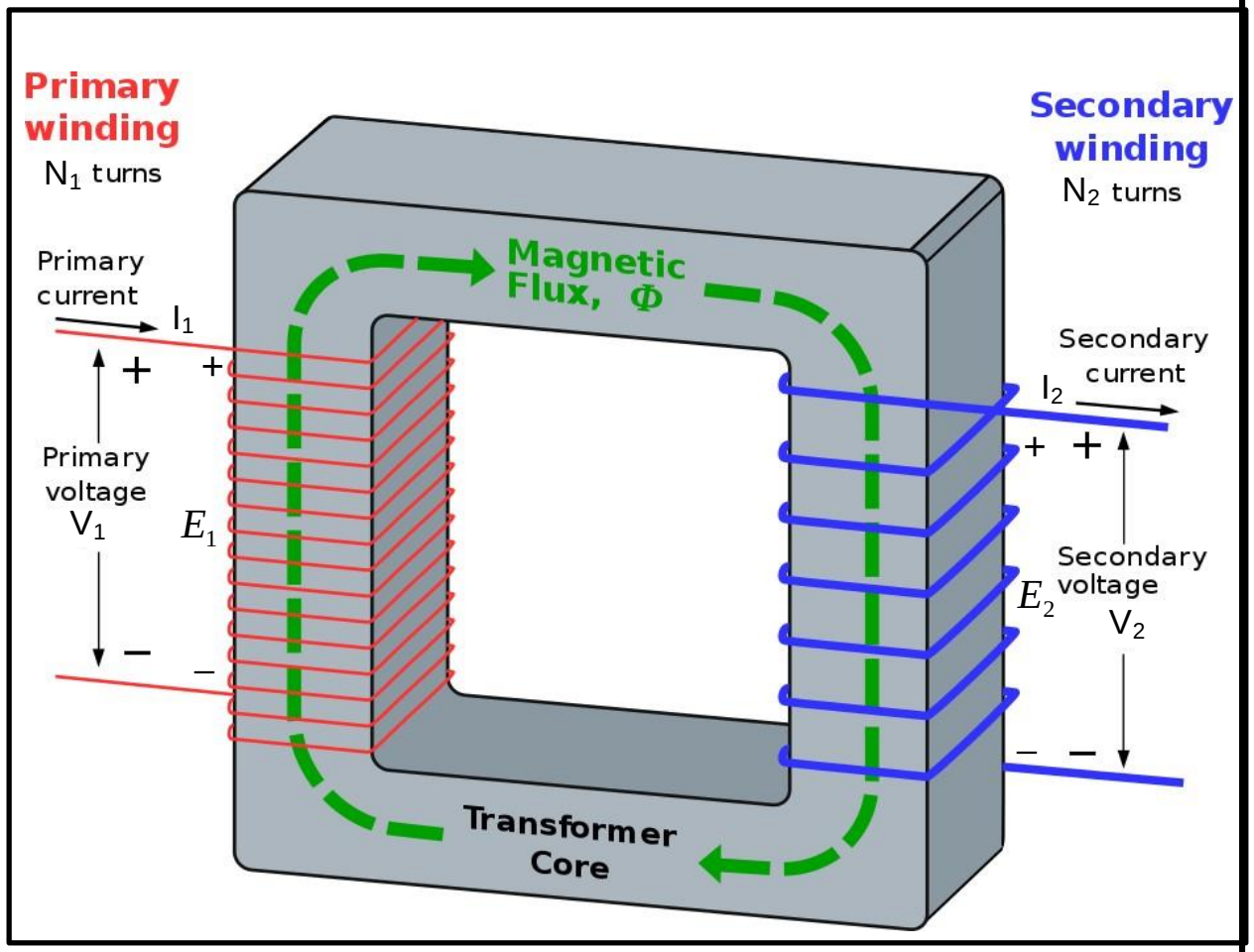
$$= \left(N_2 \frac{d\phi_{12}}{di_1} \right) \frac{di_1}{dt} = \boxed{M \frac{di_1}{dt}}$$



- This phenomenon of a **current carrying coil** (inductor) **inducing** a **voltage** across a **neighboring coil** (inductor) is '**mutual inductance**'.
- The **polarity of voltage** induced depends on the **orientation/winding** of the **two coils**, represented using "**dot convention**".

WORKING PRINCIPLE OF TRANSFORMER

- When **AC voltage** is applied to **primary** winding with **secondary open**, a "**no-load**" **current** flows in **primary** which sets up a **magnetic flux** in the core.
- This **flux** links with both primary and secondary coils => **induces EMF** by mutual (in secondary) and self (in primary) induction.



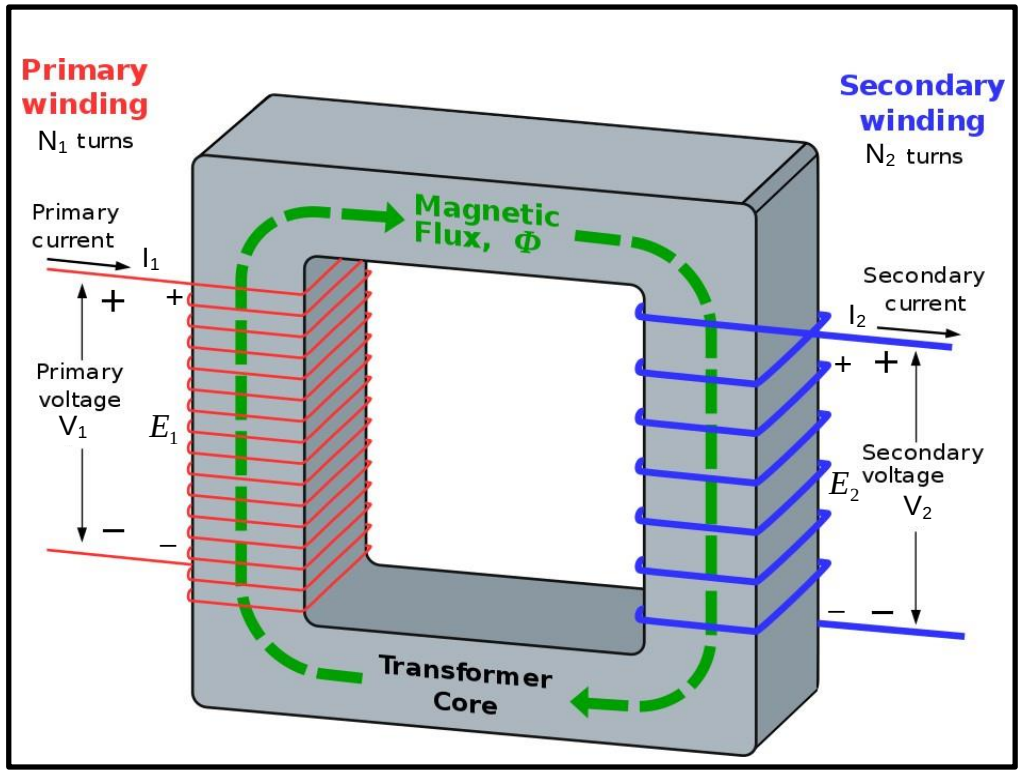
- If **secondary** is **loaded**, then **current** flows through secondary and this is **reflected** back in **primary** winding as **EMF** through mutual induction.

WORKING PRINCIPLE OF TRANSFORMER

- **Ideal transformer**: same flux ϕ linked with both windings
  => induced EMF in windings: $E_1 = N_1 \, d\phi/dt \;\; ; \;\; E_2 = N_2 \, d\phi/dt$

- **Transformation ratio**:

$$\frac{E_2}{E_1} = \frac{N_2}{N_1} = n$$



- **Lossless operation**:

$$E_1 = V_1 \;\; ; \;\; E_2 = V_2$$
$$V_1 I_1 = V_2 I_2$$

$$\Rightarrow \quad \frac{V_2}{V_1} = \frac{I_1}{I_2} = \frac{N_2}{N_1} = n$$

WORKING PRINCIPLE OF TRANSFORMER

EMF Equation:

AC voltage across primary => AC flux generated: $\phi = \phi_m \sin(\omega t)$

=> EMF induced in primary:

$$E_1 = N_1 \frac{d\phi}{dt} = N_1 \phi_m \omega \cos(\omega t) = 2\pi f N_1 \phi_m \sin\left(\omega t + \frac{\pi}{2}\right)$$ (and $E_2 = (N_2/N_1)E_1$)

$\Rightarrow$ Induced EMFs are 90^0 phase-shifted from flux

RMS values can be computed as: $E_{1_{RMS}} = 4.44 f N_1 \phi_m$

$E_{2_{RMS}} = 4.44 f N_2 \phi_m$

ISSUES IN PRACTICAL TRANSFORMER

- Primary and secondary **windings** are **not lossless**. There are **resistive losses** in **core** as well as **windings**.
- **Perfect coupling** (i.e. $k = M/\sqrt{L_1 L_2} = 1$) **not possible** due to **flux leakage** outside core and windings (typically $\boldsymbol{k < 1}$).
- **Temperature rise** due to **resistive losses** results in formation of **hot spots**, that can **damage winding** insulation and **core** properties.
- **Short circuit forces** due to **leakage flux**/magnetic field in **between** the **windings** carrying AC current. The **conductors** in the winding **exert force** on each other, leading to **mechanical stress** in winding.

NUMERICAL PROBLEMS

4. An ideal transformer has a transformation ratio of 0.125 and the primary current is 3 A when it is supplied with 240 V. Calculate the secondary voltage and current.

Soln: Transformation ratio is $$n = \frac{N_2}{N_1} = \frac{V_2}{V_1} = \frac{I_1}{I_2} = 0.125$$

The secondary voltage is $$V_2 = V_1 \times 0.125 = 30V$$

The secondary current is $$I_2 = \frac{I_1}{0.125} = 24A$$

NUMERICAL PROBLEMS

5. An ideal transformer, connected to a 240 V mains, supplies power to a 12 V, 150 W lamp. Calculate the transformation ratio and the current drawn from the mains.

Soln: The secondary current is $I_2 = \frac{P}{V_2} = \frac{150}{12} = 12.5A$

The transformation ratio is $n = \frac{V_2}{V_1} = \frac{12}{240} = 0.05$

The current drawn from supply is $I_1 = I_2 \times n = 12.5 \times 0.05 = 0.625A$

NUMERICAL PROBLEMS

6. A 5-kVA single-phase transformer has a transformation ratio of 0.1 and is fed from a 2.5-kV supply. Neglecting losses, determine (a) the full load secondary current, (b) the minimum load resistance which can be connected across the secondary winding, (c) the primary current at full load kVA.

Soln: (a) The secondary voltage is $V_2 = V_1 \times n = 2500 \times 0.1 = 250V$

=> At full load, $5kVA = V_2 \times I_2 \Rightarrow I_2 = 5000/250 = 20A$

(b) The minimum load resistance is $R_L = V_2/I_2 = 250/20 = 12.5\Omega$

(c) The primary current at full load is $I_1 = I_2.n = 20 \times 0.1 = 2\ \text{A}$

NUMERICAL PROBLEMS

7. A 100 kVA, 4000V/200V, 50 Hz single-phase ideal transformer has 100 secondary turns. Determine (a) the primary and secondary current, (b) the number of primary turns, and (c) maximum value of the flux developed in the core.

Soln: (a)

$$100KVA = V_1 \times I_1 = V_2 \times I_2$$

$$\Rightarrow I_1 = 100000/V_1 = 100000/4000 = 25A$$

$$I_2 = 100000/V_2 = 100000/200 = 500A$$

(b)

$$N_2/N_1 = V_2/V_1$$

$$\Rightarrow N_1 = (V_1/V_2)N_2 = 2000$$

(c) EMF equation for an ideal transformer is,

$$E_1 = 4.44\, f\, \phi_m N_1$$

$$\Rightarrow \phi_m = \frac{4000}{4.44 \times 50 \times 2000} = 9.01 \times 10^{-3} Wb = 9.01 mWb$$

NUMERICAL PROBLEMS

8. A single-phase, 50 Hz ideal transformer has 25 primary turns and 300 secondary turns. The cross sectional area of the core is 300 cm^2. When the primary winding is connected to a 250 V supply, determine
(a) the maximum value of the flux density in the core, and
(b) the voltage induced in the secondary winding.

Soln: (a) EMF equation for an ideal transformer is,

$$E_1 = 4.44 f \phi_m N_1 \Rightarrow \phi_m = \frac{250}{4.44 \times 50 \times 25} = 0.045 Wb$$

The maximum flux density in core is $B_m = \frac{\phi_m}{A} = \frac{0.045}{300 \times 10^{-4}} = 1.5T$

(b) EMF induced in secondary winding can be computed as,

$$N_2 / N_1 = V_2 / V_1$$

$$\Rightarrow V_2 = \left(N_2 / N_1 \right) V_1 = 3000V$$